

A Deployment Benchmark of Popular Open-Source Talking-Avatar Models Released in 2025-2026

Seven qualifying projects. One public portrait. One short narration. Six completed official pipelines measured for time, GPU memory, storage, cost, and output behavior.



Benchmark date

August 23, 2026

Author

Ethan Jiang, Tight Studio

Publication date

August 29, 2026

Artifact

[github.com/tight-studio/
open-source-talking-avatar-benchmark](https://github.com/tight-studio/open-source-talking-avatar-benchmark)

FASTEST H200

181.44 s

LiveAvatar

LOWEST VRAM

5,763 MiB

FlashHead Lite

BEST BALANCE

L40S

EchoMimicV3-Flash

Publication note. Public result MP4s are muted because the internal test narration used a macOS System Voice that cannot be publicly redistributed under Apple's license. The H.264 video tracks were copied without re-encoding.

Abstract

Talking-avatar projects claim real-time generation, identity preservation, expressive motion, and long-duration stability, but their examples use different inputs, resolutions, hardware, and timing boundaries. We evaluated every project that met two predeclared eligibility rules on August 23, 2026: an official usable release during the prior year and more than 1,000 GitHub stars. Seven projects qualified. Six produced videos; LTX-2.3 Dublt was blocked before GPU allocation because the official checkpoint required approval unavailable to the test environment.

Peak VRAM ranged from 5,763 MiB for SoulX-FlashHead Lite to 61,401 MiB for LiveAvatar. End-to-end measured generation time ranged from 181.44 seconds for LiveAvatar to 780.29 seconds for Wan2.2-S2V. Qualitative inspection found LiveAvatar to be the sharpest heavyweight result, EchoMimicV3-Flash the strongest practical balance, LongCat the most expressive, and FlashHead Lite the least memory-intensive. These are deployable official-path observations, not a controlled human-preference ranking.

1. Research question

Which recent, meaningfully adopted open-source talking-avatar projects can turn a single portrait and short narration into a convincing video, and what does each official runnable path require in time, memory, storage, and compute cost? Deployment decisions combine visual behavior with checkpoint access, dependency reliability, memory headroom, compilation, and serving cost.

2. Cohort selection

- Official usable code and weights first released from August 23, 2025 through August 23, 2026.
- More than 1,000 stars on the official GitHub repository on August 23, 2026.

Model	Stars	Release	Input	Outcome
Wan2.2-S2V-14B	17,261	2025-08-26	Image + audio	Completed
LTX-2.3 Dublt	9,226	2026-05-11	Video + text	Blocked
LongCat-Video-Avatar 1.5	7,515	2026-05-21	Image + audio	Completed
LiveAvatar	2,384	2025-12-08	Image + audio	Completed
SoulX-FlashTalk	1,476	2026-01-08	Image + audio	Completed
EchoMimicV3-Flash	1,025	2026-01-22	Image + audio	Completed
SoulX-FlashHead Lite	1,006	2026-02-12	Image + audio	Completed

GitHub stars measure attention, not quality. Projects released before the cutoff were excluded regardless of continued popularity.

3. Protocol and measurement boundary

Every successful run used the same public reference frame and the same 3.63-second sentence. Runs were deployed on Modal in model-specific CUDA containers. Each runner pins the official Git commit and checkpoint revision. Weights were already present on persistent Modal Volumes before timed generation.

The clock begins when the generation container starts and stops when the final MP4 is ready to return. It includes model loading, preprocessing, inference, decoding, and muxing, while excluding one-time internet weight download. Peak VRAM is the maximum total memory .used reported by `nvidia-smi` during the official subprocess, sampled every 0.5 seconds.

Measured fields

- Generation-function wall time and output duration.
- Peak GPU memory, output dimensions, and frame rate.
- Persistent cache size and point-in-time Modal compute estimate.
- Normalized waveform correlation between supplied and output audio.

The official pipelines use different resolutions, prompts, inference steps, quantization, and GPU classes. That is part of the deployment result, but it prevents architecture-only speed or controlled perceptual claims.

4. Quantitative results

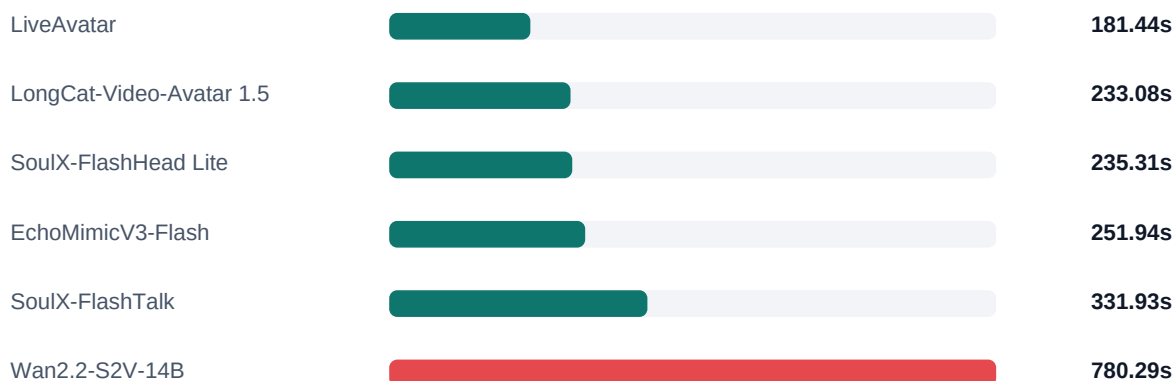
Model	GPU	Peak MiB	Time	Output	Cost	Cache
Wan2.2-S2V-14B	H200	58,681	780.29s	512x896 16 fps	\$1.3692	45.76 GiB
LTX-2.3 DubIt	-	-	Blocked	-	\$0	Gated
LongCat-Video-Avatar 1.5	H200	46,827	233.08s	480x832 25 fps	\$0.4011	44.82 GB
LiveAvatar	H200	61,401	181.44s	384x704 25 fps	\$0.3184	47.03 GiB
SoulX-FlashTalk	H200	57,805	331.93s	416x720 25 fps	\$0.5825	51.07 GiB
EchoMimicV3-Flash	L40S	33,726	251.94s	480x848 25 fps	\$0.2120	22.28 GiB
SoulX-FlashHead Lite	L40S	5,763	235.31s	512x512 25 fps	\$0.1980	7.60 GiB

Infrastructure comparison

Peak VRAM (MiB)



Measured cold-job time



FlashHead Lite is the only measured full generator that clearly falls below 16 GB. EchoMimic fits in a 48 GB class with comfortable headroom. LongCat's 46,827 MiB is operationally tight for a nominal 48 GB card. Wan, LiveAvatar, FlashTalk, and LongCat are safest on 80 GB-class hardware for these configurations.

No completed cold job generated faster than playback duration. Compilation-heavy SoulX models improve dramatically on later chunks, so warmed streaming latency should be measured separately from these end-to-end cold jobs.

5. Visual results

 Wan2.2-S2V-14B Stable identity; restrained motion; 16 fps.	 LongCat Avatar 1.5 Most expressive motion; occasional mouth exaggeration.	 LiveAvatar Sharpest heavyweight identity; restrained expression.
 SoulX-FlashTalk Stable face; clear mouth motion; costly cold compile.	 EchoMimicV3-Flash Best quality-to-infrastructure balance on L40S.	 SoulX-FlashHead Lite Lowest memory; tight crop; softer identity.

Posters are shown for print comparison. The repository contains the corresponding muted MP4s and SHA-256 checksums.

6. Model observations

Wan2.2-S2V-14B

Wan preserved identity and background with conservative facial motion. It was the slowest successful run and cost more than four times LiveAvatar for this short output. Dependency fixes were needed before inference; failed pre-denoising attempts are excluded.

LongCat-Video-Avatar 1.5

The eight-step INT8 path produced the most visible performance, including a hand gesture absent from the reference image. It ran on one H200 with context parallelism set to one. Peak allocation leaves little 48 GB operational headroom.

LiveAvatar

The four-step FP8 path produced the sharpest identity, convincing mouth shapes, blinking, and restrained expression. It was the fastest H200 job but still needed 61,401 MiB and model offloading during final VAE decode.

SoulX-FlashTalk

FlashTalk was visually stable and designed for continuous chunks. First-chunk graph compilation took 119.58 seconds; later chunks took about 3.75 seconds. The cache was the largest in the test.

EchoMimicV3-Flash

The 1.3B eight-step path produced clean 25 fps motion on L40S. Its observed 33,726 MiB peak is for the official 768-area configuration, not a theoretical minimum-memory mode.

SoulX-FlashHead Lite

FlashHead used 5,763 MiB. First compilation took 155.7 seconds, while later chunks took about 0.19 seconds. The close crop and weaker mouth motion reduced visual polish.

LTX-2.3 Dublt

The official checkpoint required approval unavailable to the test environment. The job stopped during CPU-only weight preparation. No unofficial mirror was substituted and no GPU was allocated.

7. Voice behavior

The six completed image-and-audio systems animate supplied speech and mux that speech into their output; they do not select or clone a voice. Correlation with the aligned input ranged from 0.999198 to 0.999897. Small differences are consistent with resampling and AAC encoding rather than creation of a new speaker.

Model	Input speech	Voice learned?	Correlation
Wan2.2-S2V-14B	Preserved	No	0.999897
LTX-2.3 DubIt	Target text	From reference video	Not run
LongCat-Video-Avatar 1.5	Preserved	No	0.999562
LiveAvatar	Preserved	No	0.999897
SoulX-FlashTalk	Preserved	No	0.999751
EchoMimicV3-Flash	Preserved	No	0.999198
SoulX-FlashHead Lite	Preserved	No	0.999751

DubIt is a different interface class: it takes target text and a voiced reference video, aiming to retain vocal identity while synthesizing new speech.

8. Deployment interpretation

- LiveAvatar for the sharpest observed heavyweight output when 80 GB-class GPU capacity is available.
- EchoMimicV3-Flash for the best observed balance of result quality, cache size, and 48 GB-class deployment.
- LongCat Avatar 1.5 when expressive body motion matters more than conservative motion.
- SoulX-FlashHead Lite when low memory and warmed streaming speed dominate visual polish.

9. Limitations

- One portrait and one short English sentence; no demographic, language, emotion, occlusion, profile-view, or long-duration coverage.
- No preregistered blinded human study; qualitative judgments are author observations.
- Official paths differ in resolution, frame rate, steps, quantization, prompts, and hardware.
- Cold-job timing disadvantages compilation-heavy streaming systems.
- A 0.5-second VRAM poll can miss shorter allocation spikes.
- Modal prices and scheduling are point-in-time cloud observations, not universal total cost of ownership.
- Normalized summaries are public, but full internal job logs, cached weights, and the source narration waveform are not.
- Access, licenses, stars, dependencies, and upstream recommendations can change.

10. Reproducibility package

The repository contains exact runner code, pinned upstream commits and checkpoint revisions, CSV and JSON results, SHA-256 checksums, the public reference frame, muted outputs, poster frames, and build/validation scripts. The reproduction guide defines every measurement boundary and provides all Modal commands.

A stronger follow-up should repeat multiple seeds, use diverse consented portraits and rights-cleared multilingual voices, separate cold-start from warmed throughput, record failure rate, and run a preregistered blinded perceptual study. It should be published as a new dated dataset rather than overwrite this cohort.

11. Conclusion

Recent open-source talking-avatar pipelines can generate convincing short portrait videos from a single image, but deployment requirements differ by an order of magnitude. LiveAvatar led heavyweight visual quality in this test. EchoMimic came close while using a cheaper GPU and less than half the model storage of the large generators. FlashHead showed that a complete generator can fit well below 16 GB, with visible quality tradeoffs. For serving decisions, checkpoint size, memory headroom, warm-up behavior, and access terms matter as much as the strongest sample frame.

References

1. Wan-Video, Wan2.2.
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5. Soul AI Lab, SoulX-FlashTalk and SoulX-FlashHead.
6. Ant Group, EchoMimicV3.
7. Pexels, reference video by AI25.Studio and Pexels License.
8. Modal, pricing observed August 23, 2026.
9. Apple, macOS Software License Agreement, System Voices section.

Repository: github.com/tight-studio/open-source-talking-avatar-benchmark